Predictive Sensorimotor Grasping (PSG)

System architecture for MAGPIE AutoGrasp: predict the contact, compare at control points, learn value from success

Adam Abid (ACWA) · Correll Lab, CU Boulder · technical note, draft of October 8, 2026

Summary. AutoGrasp today chooses a grip force before contact and then reacts to slip; nothing it feels during one grasp changes the next. PSG adds the piece human grasping relies on: a forward model that predicts what the grasp should feel like (the load at lift-off and the grip-to-load ratio at which the object slips), a comparator that checks that prediction at a few contact events and corrects the grip on the spot, and a per-object correction that fixes the prediction after a single grasp without touching any trained weights. A small value head learns the safety margin from success alone. The VLA, GraspGenX and DeliGrasp stay as they are. In a MuJoCo comparison on objects whose appearance misrepresents their weight, PSG lifted and held 97% of grasps versus 39% for a DeliGrasp-style semantic prior and 51% for Exp-Force-style retrieval, and its corrections did not leak between identical-looking objects. The method is integrated behind a flag in `deligrasp_node` of `magpie_control_rosify` and has not yet run on the UR5.

1 Why: robots react, humans predict

Before the fingers touch an object, a person already expects how heavy and slippery it is. Contact, lift-off and slip are checkpoints where that expectation meets sensation. When an object is heavier than it looked, the mismatch triggers a corrective increase in grip within about 70 ms and rewrites the expected weight in a single lift [1]. The motor system does this even while perception stays fooled: after twenty lifts of a large and a small cube of equal weight, people still judge the small one heavier, yet their grip forces are correct after one or two lifts [2].

Recent robot work adds experience in other ways. Exp-Force retrieves similar past grasps to choose a pre-grasp force [7]; OmniVIC retrieves experience to set impedance [8]; RL-Token learns a residual on a frozen VLA from reward [9]; PhysMem keeps verified rules for a VLM planner [11]; MemoryVLA stores tokens inside the policy [12]. None of them predicts the sensory signature of the grasp and checks it mid-grasp, and none separates the surprise signal from the reward signal. That separation is the organising idea of PSG.

2 Design principles

- **Three learners, three error signals.** Following Doya's account of the cerebellum, basal ganglia and cortex [4]: a forward model learns from sensory prediction error (supervised), a value head learns from reward (reinforcement), and the visual representation is pretrained and frozen. Prediction error is never used as reward.
- **Two speeds of memory.** Shared model weights change slowly and only offline; a small per-object correction changes after one grasp and touches nothing else, the fast/slow split of Complementary Learning Systems [5]. Nothing learned about one object can overwrite another.
- **Leave the stack alone.** Perception, GraspGenX, Gemini arbitration and DeliGrasp's closing loop are unchanged. PSG sets DeliGrasp's starting force and turns the retreat into a measured lift.

- **Everything logged is a label.** Every grasp writes its prediction, each control point and the outcome. The next training set accumulates while the robot works.

3 Architecture overview

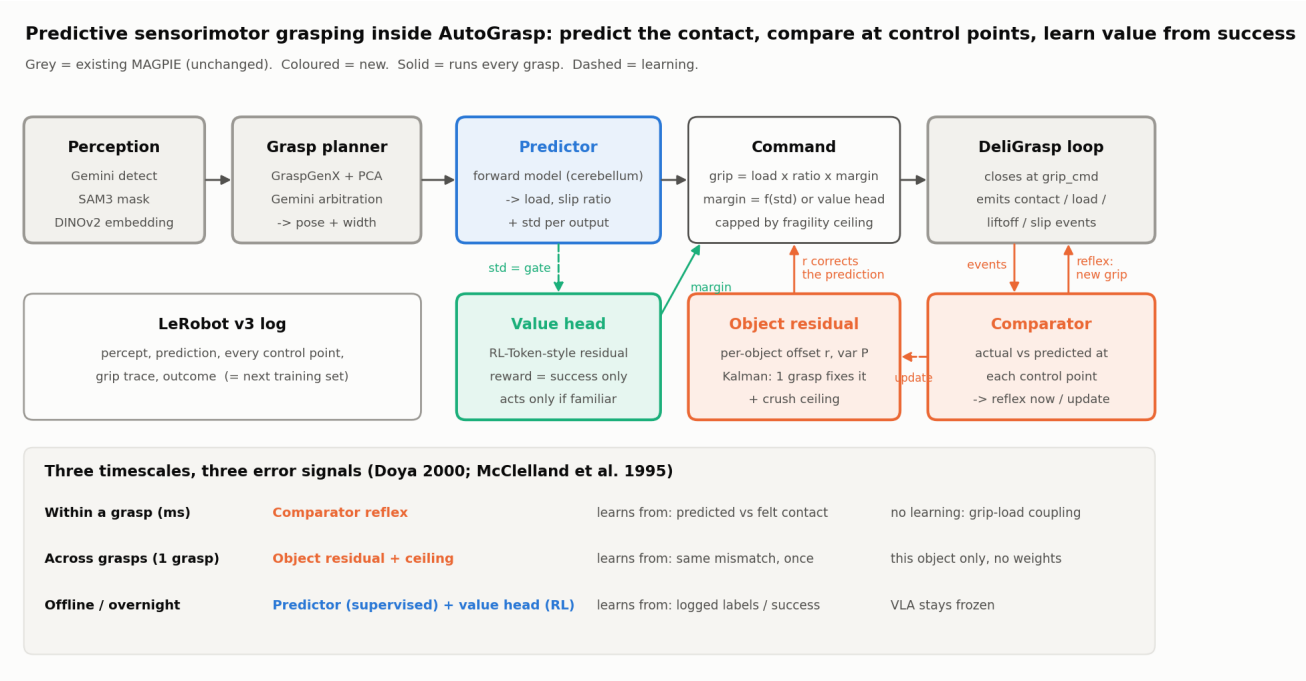


Figure 1. PSG inside AutoGrasp. Grey boxes are the existing MAGPIE stack; coloured boxes are new. The predictor (blue) sets the commanded grip; the comparator and object residual (orange) correct it within a grasp and across grasps; the value head (green) chooses the safety margin when the predictor is confident. The lower band lists the three timescales.

4 Components

4.1 Percept

The input is a DINOv2 ViT-S/14 embedding of the detection crop (384 values) plus a short geometric descriptor: object width and height in metres from the bounding box and depth, and the depth itself. Without torch the node falls back to a colour histogram; the model must be trained and run with the same embedder, which the log records. The object instance key (`object_id`) is separate from appearance on purpose: two identical bottles look the same but must be learned separately, as a person tells them apart by where they sit.

4.2 Predictor (forward model)

The predictor maps the percept x to two quantities in log space, each with a standard deviation: the vertical load the fingers carry at lift-off, L (newtons, about $m(g + a)$), and the grip-to-load ratio at which the object slips, ρ (about $1/2\mu$ for two frictional contacts). It is a bootstrap ensemble of eight ridge regressors on 512 random Fourier features. The ensemble spread gives a variance that grows away from the training data and an out-of-bag noise floor is added, so

$$\mu(x) = \text{mean}_k\, f_k(x), \qquad \sigma^2(x) = \text{var}_k\, f_k(x) + \sigma^2_{\text{noise}}$$

It trains in milliseconds on a few hundred grasps and is never updated online. A heteroscedastic network can replace it once data grows; only `predict()` has to keep its signature.

4.3 Command

The commanded grip is the predicted need times a safety margin that grows with uncertainty, as people grip harder when unsure, capped by any fragility ceiling learned for this object:

$$g_{\text{cmd}} = \min(\, \exp(\mu_L + \mu_\rho) \cdot m,\, g_{\text{ceil}}\,), \qquad m = 1.25 \cdot \exp(\|\sigma\|) \quad \text{or} \quad m = \text{value head}$$

4.4 Comparator and control points

The comparator is a callback that the lift calls at every control tick. It answers two questions separately: should the grip change now (reflex), and should this object's prediction change (update).

Control point	What is checked	Reflex (now)	Learning (after)
Contact	both fingers loaded	none	none
Load ramp (before lift-off)	felt load vs predicted + 2σ	grip = load · $\hat{\rho}$ · 1.3 (grip-load coupling)	none
Lift-off	settled load, z = (log L − μ_L)/σ_L	none	if |z| > 1: Kalman update of r_L
Slip	load falls below 0.6 × lift-off load	grip × 1.4	r_ρ raised to at least the ratio it slipped at
Crush (operator label)	/psg/mark_crushed	none	ceiling = 0.7 × max grip used

4.5 Object residual (one-grasp correction)

Each object carries a residual r for each predicted quantity with variance P, starting at $r = 0$ and $P = \sigma^2(x)$. The prediction used is $\mu(x) + r$. At lift-off the innovation $e = \log L_{\text{felt}} - (\mu_L + r_L)$ drives a scalar Kalman step with sensor noise R:

$$K = P \,/\, (P + R), \qquad r \leftarrow r + K\,e, \qquad P \leftarrow (1 - K)\,P$$

When the predictor was unsure, K is close to 1 and one grasp fixes the next prediction. When it was confident but badly wrong, which is exactly the deceptive-object case, a plain Kalman step would correct only part of the error. PSG treats a surprise beyond 3σ as evidence that the prior itself is wrong for this object (unexpected uncertainty, [6]) and resets $P = e^2$ before the step, so the gain is again near 1. Residuals are saved to `residuals.json` after every grasp.

4.6 Value head

The value head picks the margin m from {1.05, 1.2, 1.4, 1.7, 2.1} using a LinUCB contextual bandit over [1, μ_L, μ_ρ, ‖σ‖]. Reward is 1 for a lift-and-hold minus 0.05(m − 1); prediction error never enters it. A gate lets it act only when ‖σ‖ < 0.35, so on unfamiliar objects the uncertainty-scaled default margin applies. It is warm-started offline from logged grasps. The planned robot version is an RL-Token actor-critic on the frozen VLA's readout token [9] with the same `choose/update` interface.

4.7 Instrumented lift

The retreat after DeliGrasp becomes four phases, implemented ROS-free in `psg/lift.py`: (1) tare the wrist force sensor with the object still on the table, so squeeze forces and gripper weight cancel and the felt load is $|\mathbf{F} - \mathbf{F}_{\text{tare}}|$; (2) lift 2 cm at 1 cm/s, emitting a load event every 20 ms and a lift-off event with the settled load at the end; (3) the rest of the retreat plus a 1.5 s hold, watching for slip or drop; (4) optionally, during data collection only, step the grip down until the load falls to measure the slip ratio label.

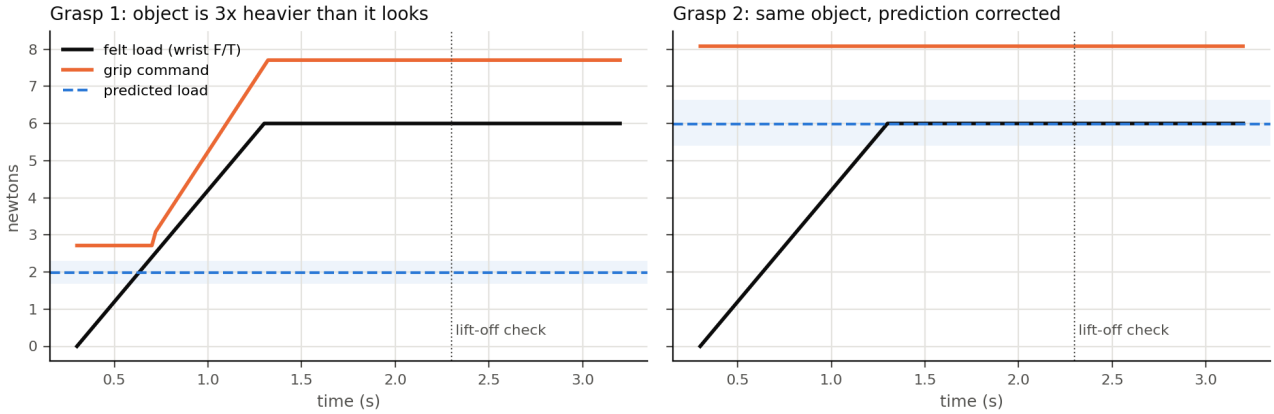


Figure 2. One object, two grasps, produced by the real PSG code against a simulated wrist sensor. Grasp 1: the object is three times heavier than predicted; as the felt load passes the predicted band the comparator couples grip to load and the lift succeeds; lift-off triggers the residual update. Grasp 2: the corrected prediction matches the felt load and the grip is set right from the start.

5 Integration in the MAGPIE ROS 2 stack

PSG lives in `magpie_control/psg/` on branch `psg-predictive-grasp` of `magpie_control_rosify`. `deligrasp_node` runs it when `use_psg:=true`; with the flag off the node is unchanged.

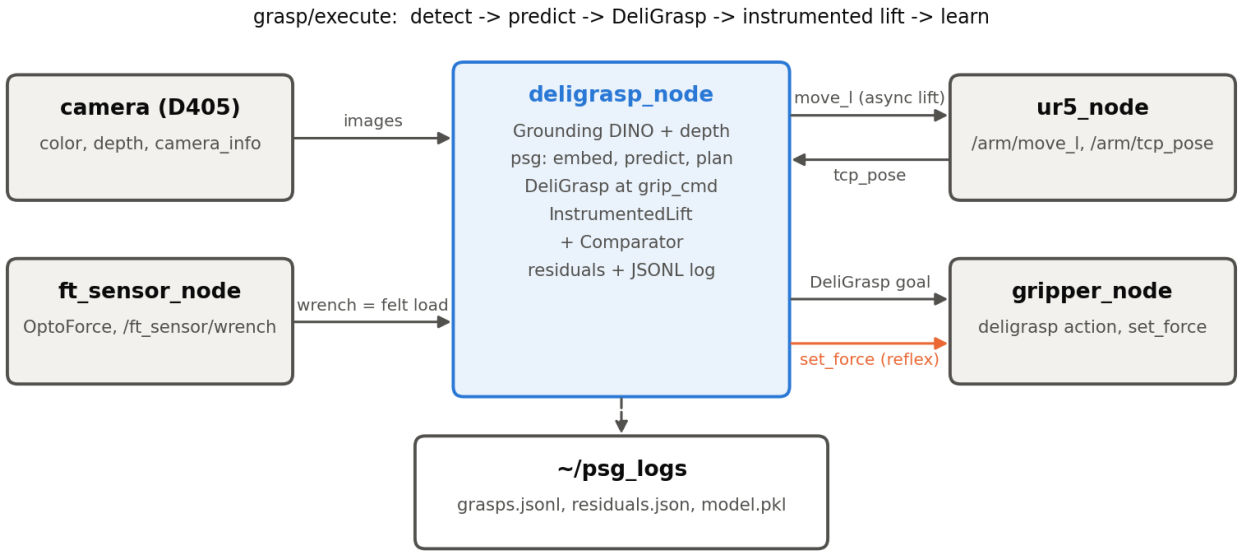


Figure 3. Data flow per grasp. The reflex reaches the gripper through /gripper/set_force; the lift uses asynchronous move_l with completion read from /arm/tcp_pose; the felt load comes from the OptoForce.

Parameter	Default	Purpose
use_psg	false	off = stock DeliGrasp
psg_mode	predictor+update	predictor | predictor+update | full | full-nogate (ablation ladder)
psg_model_path	(empty)	empty = collect mode: fixed initial force, everything logged
psg_log_path / psg_residual_path	~/psg_logs/…	training log (JSONL) and per-object residuals
psg_embedder	dinov2	dinov2 (needs torch) or hist fallback
psg_probe	false	measure slip-ratio labels after a good hold (drops the object)
object_id	(object_query)	instance key for the per-object residual
wrench_topic	/ft_sensor/wrench	wrist F/T topic

Each grasp appends one JSON line: embedding and embedder, grasp descriptor, commanded grip, prediction and its standard deviation, margin, success, slip, drop, lift-off load, slip ratio if probed, maximum grip, number of reflexes, every control point and the full load/grip trace.

6 Data and training pipeline

Phase	What runs	Output	Done when
0 Collect	use_psg:=true, no model, psg_probe:=true; 10–15 objects × ~5 grasps at varied initial_force	grasps.jsonl	≥150 lifted grasps, ≥20 probed
1 Train	ros2 run magpie_control psg_train	model.pkl + calibration printout	≈68% of held-out errors within 1σ
2 Predict + update	psg_mode:=predictor+update	residuals.json, new log rows	deceptive-object test (§8) passes
3 Value head	psg_mode:=full; swap bandit for RL-Token	margin policy	beats fixed margin on fragile + slippery set

7 Evaluation so far (MuJoCo)

A floating parallel-jaw gripper with force-controlled fingers lifts boxes 10 cm and holds them for 1 s. Each object has a true mass, friction coefficient and crush limit; the appearance vector is a fixed random projection of size and material plus noise, standing in for DINOv2. Training objects are honest (density follows material). The test set adds 30 unseen honest objects and six deceptive ones: a big hollow box and a small dense block, identical bottles empty and full, hollow metal that dents, and lead-filled plastic. Baselines receive the same slip reflex so only prediction and learning differ: a DeliGrasp-like semantic prior (mass and friction from the material label, margin 1.5) and an Exp-Force-like retrieval (appearance kNN over successful forces, seeded with the same Phase-0 data and growing online). Three seeds.

Method	Deceptive: lifted & held	Deceptive: grip appropriate	Unseen: force error (N)	Unseen: grip appropriate
DeliGrasp-like (semantic prior)	39%	0%	1.93	92%
Exp-Force-like (retrieval)	51%	16%	3.07	44%
PSG: predictor only	83%	0%	0.99	97%
PSG: predictor + one-shot update	97%	78%	0.93	97%

Method	Deceptive: lifted & held	Deceptive: grip appropriate	Unseen: force error (N)	Unseen: grip appropriate
PSG: full (value head, gated)	97%	73%	0.70	91%
PSG: full, no gate	97%	73%	0.78	90%

“Appropriate” means the initial grip was between 1× and 2.5× what the object needed and below its crush limit.

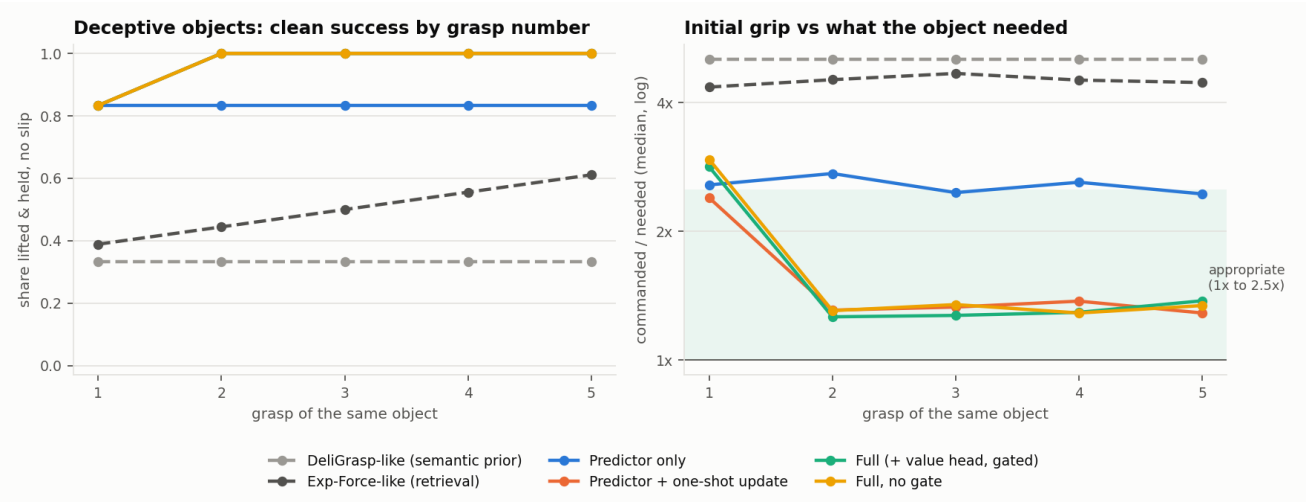


Figure 4. Deceptive objects, five interleaved grasps each. With the one-shot update, clean success reaches 100% from grasp 2 and the initial grip falls from about 2.5× to about 1.3× of need. The predictor alone and the semantic prior do not improve; retrieval improves slowly.

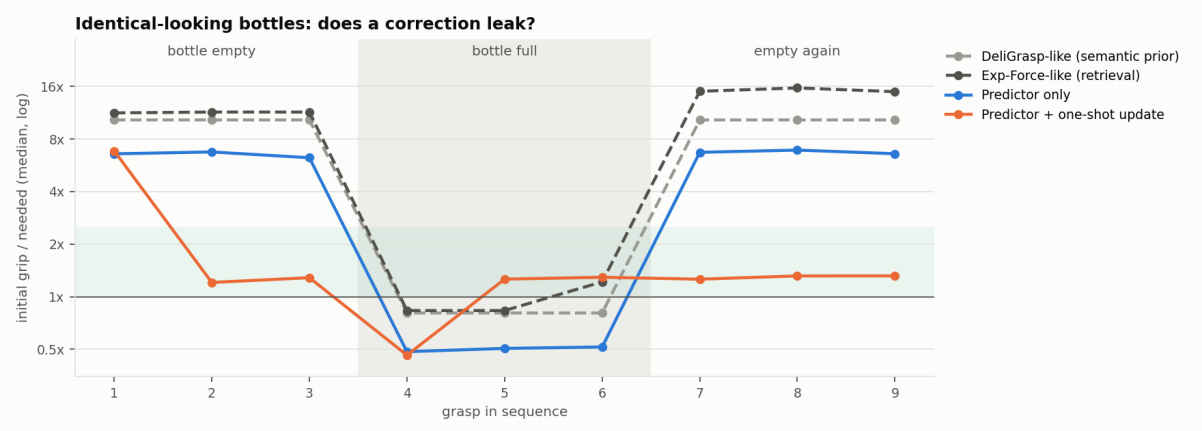


Figure 5. Identical-looking bottles: empty ×3, full ×3, empty ×3. After the full bottle, appearance retrieval grips the empty one at about 15× need; PSG's per-object residual stays near 1.3×.

What the simulation supports: the one-shot update corrects deceptive objects by the second grasp without leaking to look-alikes, and the prediction-triggered load reflex already lets the predictor alone beat both baselines on first contact (83% vs 39–51%). What it does not support: the value head and its gate add nothing measurable here (97% either way). The simulation has one control knob and a simple failure model, so there is little margin policy to learn. The value-head claim should be tested on hardware with fragile and slippery objects or dropped.

8 Planned hardware experiments

- **Size-weight pair:** a large empty box and a small box filled with coins, same colour; five interleaved grasps each. Prediction: load-prediction error and grip ratio collapse by grasp 2 with predictor+update and stay flat with the predictor alone and with stock DeliGrasp.
- **Look-alikes:** two identical bottles, one full, with different object_id values; the A-A-A, B-B-B, A-A-A sequence tests that a correction does not leak.
- **Exp-Force's own metric:** force error and share of appropriate forces on about 30 unseen objects, for comparison with their reported 0.43 N best-case error and 63%→87% appropriate force [7].
- **Ablation ladder:** stock DeliGrasp, predictor, predictor+update, full, full-nogate; success, slips, reflexes, force error.
- **Dual-belief benchmark (stretch):** keep the VLM's perceptual belief ('looks heavy') separate from the sensorimotor belief and act only on the latter, as people do in the size-weight illusion [2]. Score three curves over repeated grasps: motor error (should drop by grasp 2), perceptual estimate (allowed to stay wrong) and the gap between them (should become a stable, object-specific constant).

9 Limitations and open questions

- Simulation only so far: boxes, a stand-in embedding, one control knob. No result here has run on the UR5.
- The felt load requires the wrist OptoForce. Without it there is no lift-off control point and no load reflex.
- Instance identity is supplied (object_id). Truly indistinguishable objects without tracking would fall back to the last lift, as a person would; that is a fair additional experiment, not yet built.
- Crushes are labelled by hand; there is no deformation sensing.
- The reflex runs on a 50 Hz force stream plus a ROS service call: tens of milliseconds, adequate for a slow lift and untested on fast slips.
- Phase-0 friction labels need the probe, which drops the object; collection needs a soft landing area.

10 Relation to prior work

Work	What it learns	Per-object, one grasp?	Corrects mid-grasp?	Separates surprise from reward?
Exp-Force [7]	pre-grasp force by retrieval + VLM	no	no	no value function
OmniVIC [8]	impedance via retrieval + in-context learning	no	reactive only	no value function
RL-Token [9]	residual actor-critic on frozen VLA	no	no	reward only
RECALL [10]	recovery demos at uncertain states + replay	no	no	uncertainty, no reward
PhysMem [11]	verified rules for a VLM planner	per rule	no	surprise triggers consolidation
MemoryVLA [12]	token memory inside the VLA	no	no	neither
PSG (this note)	contact signature + variance; per-object residual; margin	yes	yes, at control points	yes, by design

References

[1] Flanagan, J. R., Bowman, M. C., & Johansson, R. S. (2006). Control strategies in object manipulation tasks. *Current Opinion in Neurobiology*, 16(6).

[2] Flanagan, J. R., & Beltzner, M. A. (2000). Independence of perceptual and sensorimotor predictions in the size–weight illusion. *Nature Neuroscience*, 3(7).

[3] Johansson, R. S., & Westling, G. (1984). Roles of glabrous skin receptors and sensorimotor memory in automatic control of precision grip. *Experimental Brain Research*, 56.

[4] Doya, K. (2000). Complementary roles of basal ganglia and cerebellum in learning and motor control. *Current Opinion in Neurobiology*, 10(6).

[5] McClelland, J. L., McNaughton, B. L., & O'Reilly, R. C. (1995). Why there are complementary learning systems in the hippocampus and neocortex. *Psychological Review*, 102(3).

[6] Yu, A. J., & Dayan, P. (2005). Uncertainty, neuromodulation, and attention. *Neuron*, 46(4).

[7] Exp-Force: experience-conditioned pre-grasp force selection. arXiv:2603.08668 (2026).

[8] OmniVIC: a self-improving variable impedance controller. arXiv:2510.17150 (2025).

[9] RL Token: bootstrapping online RL with vision-language-action models. Physical Intelligence, arXiv:2604.23073 (2026).

[10] RECALL: continual learning for VLAs via uncertainty-guided recovery. arXiv:2606.23617 (2026).

[11] Li, H., You, Y., Su, H., & Guibas, L. PhysMem: self-evolving physical memory for robot manipulation. arXiv:2602.20323 (2026).

[12] Shi, H., et al. MemoryVLA: perceptual-cognitive memory in vision-language-action models. ICLR 2026, arXiv:2508.19236.

[13] Wolpert, D. M., & Kawato, M. (1998). Multiple paired forward and inverse models for motor control. *Neural Networks*, 11.

[14] DeWolf, T., Stewart, T. C., Slotine, J.-J., & Eliasmith, C. (2016). A spiking neural model of adaptive arm control. *Proc. R. Soc. B*, 283.